

PRODUCT DATA SHEET

Decorative HPL (HGS/VGS)



DESCRIPTION

Decorative HPL is a high pressure laminate manufactured from layers of phenol impregnated Kraft papers and a melamine impregnated decorative surface paper, consolidated under high pressure and temperature in accordance with EN 438, offering a wide range of decorative designs, colors, patterns, and surface finishes to enhance the visual appearance of interior furniture and building applications.

Description: Decorative HPL

Laminate Grade 1: HGS (Horizontal General Purpose Standard)

Laminate Grade 2: VGS (Vertical General Purpose Standard)

Size: 1220x2440mm, 1220x2800mm, 1300x2800mm, 1220x3050mm, 1300x3050mm

Thickness: 0.5 to 1.80 mm

APPLICATION / APPLICATION AREA

For use as front and top facing surfaces on furniture substrates, providing decorative aesthetics and surface protection in line with contemporary design trends



QUALITY CHARACTERISTICS / TECHNICAL DATA

Decorative HPL in both Horizontal and Vertical grades (HGS/VGS) are classified in accordance with EN 438-3 and tested as per EN 438-2 to evaluate their performance, durability, and resistance to various physical and environmental conditions.

PROPERTIES	TEST METHOD	MINIMUM REQUIREMENT ACCORDING TO EN 438-3:2016-02, TABLES 4 and 5	RESULTS HGS/VGS
Resistance to surface wear	EN 438-2:2016-02, p. 10	Revolutions: > 150 until IP	> 400
Resistance to immersion in boiling water	EN 438-2:2016-02, p. 12	Appearance Gloss finish $\geq$ Grade 3 Other finishes $\geq$ Grade 4	Grade 5 Grade 5
Resistance to water vapour	EN 438-2:2016-02, p. 14	Appearance Gloss finish $\geq$ Grade 3 Other finishes $\geq$ Grade 4	Grade 4 Grade 4
Resistance to dry heat (160°C)	EN 438-2:2016-02, p.16	Appearance Gloss finish $\geq$ Grade 3 Other finishes $\geq$ Grade 4	Grade 4 Grade 4
Dimensional stability at elevated temperature	EN 438-2:2016-02, p.17	Cumulative dimensional change Longitudinal L: 0.55% (max) Transversal T: 1.05% (max)	0.4 0.7
Resistance to wet heat (100°C)	EN 438-2:2016-02, p.18	Appearance Gloss finish $\geq$ Grade 3 Other finishes $\geq$ Grade 4	Grade 5 Grade 5
Resistance to impact with a small diameter ball	EN 438-2:2016-02, p.20	Spring force $\geq$ 20 N	> 33
Resistance to impact with a large diameter ball	EN 438-2:2016-02, p.21	Drop height $\geq$ 800 mm Indentation diameter $\leq$ 10 mm	> 1800 9
Susceptibility to cracking (laminated thickness < 2 mm)	EN 438-2:2016-02, p.23	Appearance $\geq$ Grade 4	Grade 5
Resistance to scratching	EN 438-2:2016-02, p.25	Force (min.) smooth surface $\geq$ Grade 3 structured surface $\geq$ Grade 4	Grade 3 Grade 4
Resistance to staining	EN 438-2:2016-02, p. 26	Appearance Group 1: = Grade 5 Group 2: = Grade 5 Group 3: $\geq$ Grade 4	Grade 5 Grade 5 Grade 4
Light fastness	EN 438-2:2016-02, p. 27	Contrast Grey Scale: 4 to 5	4 to 5
Density	EN 323:1993	$\geq 1.35 \text{ g/cm}^3$	1.39

Rating Scale Interpretation (EN 438):

Grade 5 no visible change

Grade 4 slight change of gloss and/or colour, only visible at certain viewing angles

Grade 3 moderate change of gloss and/or colour

Grade 2 marked change of gloss and/or colour

Grade 1 blistering and/or delamination

DIMENSIONAL PROPERTIES

Dimensional properties of Decorative HPL are assessed in accordance with EN 438-2 to ensure accuracy in thickness, size, edge straightness, squareness, and flatness.

PROPERTY	TEST METHOD	ATTRIBUTES OF PERFORMANCE	RESULTS (AS PER EN 438-3)
Thickness	EN 438-2:2016-02, p. 05	Thickness tolerance	$0.5 \leq T < 1 : \pm 0.1$
Size	EN 438-2:2016-02, p. 06	Length and width	+ 10 / - 0
Straightness of edges	EN 438-2:2016-02, p. 07	Straightness of edges	≤ 1.5
Squareness	EN 438-2:2016-02, p. 08	Squareness	≤ 1.5
Flatness	EN 438-2:2016-02, p. 09	Flatness (measured on full-size sheet)	60

SURFACE FINISH

The surface finish test is conducted in accordance with EN 13722 to evaluate the angle of light reflection for each surface finish. The results are expressed in gloss units (GU), providing a precise measure of the reflective characteristics.

SURFACE	MAXIMUM DEVIATION	GLOSS UNITS
High Gloss Surface	± 15	> 70
Semi-Gloss Surface	± 10	30 - 70
Suede Surface	± 5	10 - 30
Matt surface	± 3	< 10